

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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BIOLOGY

Paper 2 Structured Questions

9744/02

14 September 2018

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **ALL** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Do not open this booklet until you are told to do so.

For Examiner's Use	
1	/ 9
2	/ 10
3	/ 12
4	/ 8
5	/ 9
6	/ 12
7	/ 10
8	/ 11
9	/ 8
10	/ 6
11	/ 5
Total	/ 100

Answer **all** questions in this section.

- 1 Collagen is the main structural protein in the human body. It strengthens the tendons and supports the skin and internal organs. Fig.1.1 shows the organization of collagen fibres.

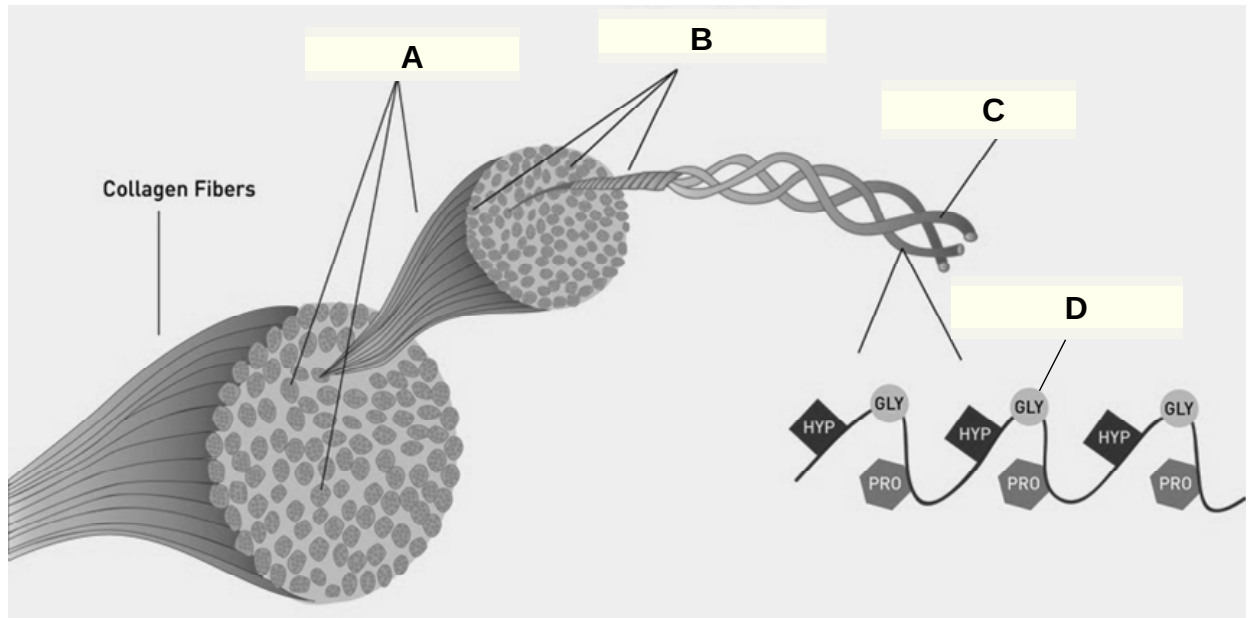


Fig. 1.1

- (a) Label structures **A**, **B**, **C** and **D** in Fig. 1.1. [2]

A: Collagen fibrils;
B: Tropocollagen / Collagen triple helix;
C: Loose helical polypeptide chain;
D: Glycine;

- (b) With reference to Fig. 1.1, describe the bonds involved in the formation of collagen which contribute to its high tensile strength. [2]

a. Within 1 tropocollagen molecule numerous hydrogen bonds* form between amino acids of adjacent polypeptide chains;;
b. covalent cross-links* form between lysine* residues at C and N ends of adjacent/parallel tropocollagen molecules;;

Ignore if mention staggered arrangement

- (c) Suggest why the assembly of collagen takes place outside the cell. [2]

a. Cleaving of the ends of the tropocollagen before assembly into collagen fibres is performed by enzymes found only outside of the cell ;;
b. Collagen molecule is too large to pass through the cell surface membrane and has to be assembled outside the cell ;;

- c. Assembly of collagen outside the cell allows alignment of microfibrils / formation of bonds between the microfibrils ;;

Max 2

- (d) Certain pathogenic bacteria such as *Clostridium histolyticum*, have collagenases which digests collagen tissue of their hosts, causing a form of tissue death known as gangrene.

The action of collagenase can be seen in Fig.1.2, illustrating the specificity of the active site in binding to a segment of collagen and eventually cleaving it into 2 segments.

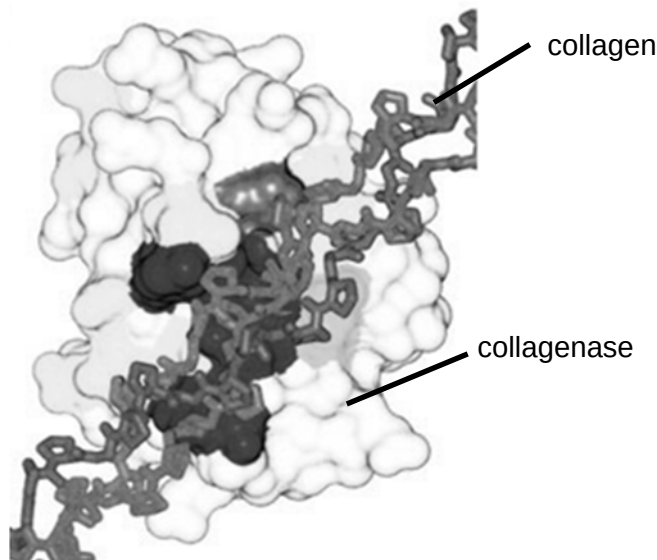


Fig. 1.2

- (i) Explain the function of amino acid residues situated at the active site of collagenase. [2]

- Catalytic residues act on bonds in substrate and help to catalyse conversion of substrate to product;;
- Amino acids in collagenase have specific R groups* which facilitates cleaving of peptide bonds e.g. acid-base catalysis;;
- Contact residues help to hold substrate at correct orientation/position;;
- via weak interactions such as hydrogen bonds, ionic bonds & hydrophobic interactions;;

Max 2

- (ii) A scar is an area of fibrous tissue that replaces normal skin after an injury. All scarring is composed of the same collagen as the tissue it has replaced, but the composition of the scar tissue, compared to the normal tissue, has slightly different arrangement. Scar tissue also lacks elasticity unlike normal tissue which distributes fiber elasticity. The extend of scarring depends on the amounts of collagen expressed at the injury site.

Santyl® Ointment is an enzymatic ointment which contains collagenase. The enzyme collagenase is derived from *Clostridium histolyticum*.

Suggest a therapeutic use of collagenase. [1]

- a. Collagenase can be applied to remove callouses/scar/warts;;

[Total: 9]

- 2 Fig. 2.1 shows an incomplete diagram of the fluid mosaic model of membrane structure. The diagram shows the cell surface membrane of a eukaryotic cell.

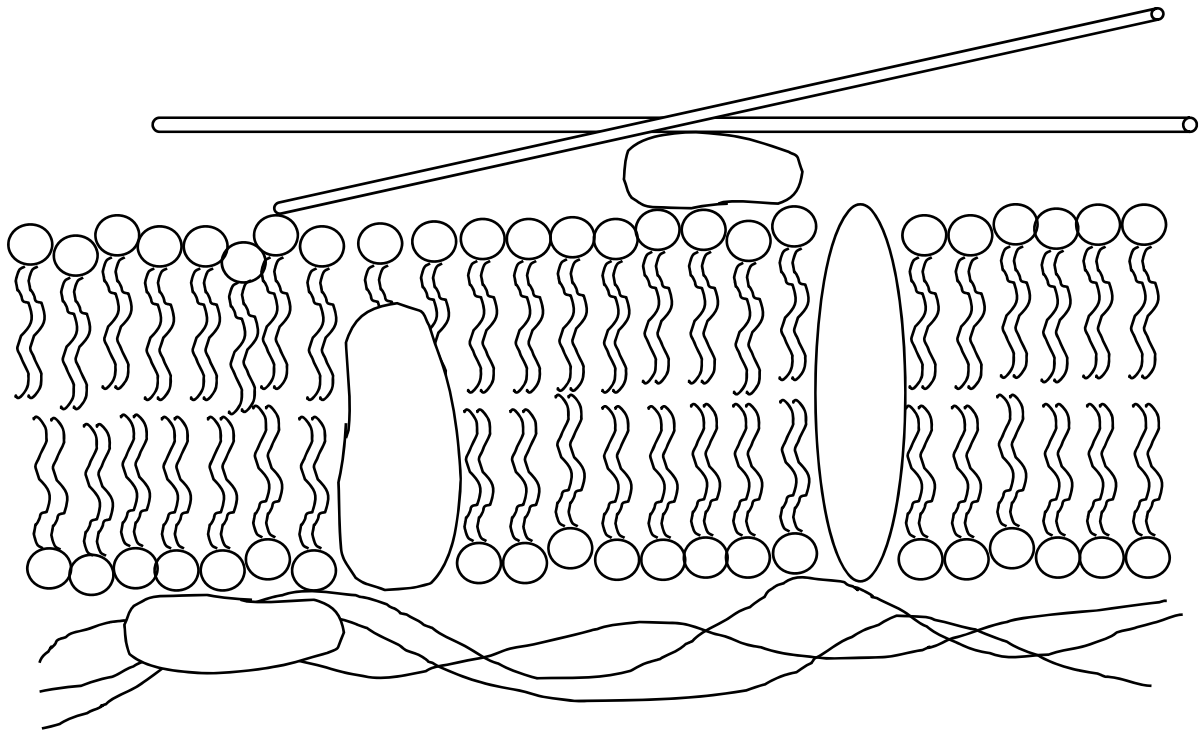


Fig. 2.1

- (a) State what is meant by fluid mosaic model. [2]

fluid

- a. refers to phospholipids (and proteins), move / constant motion / diffuse laterally and rotate on their axis, due to weak hydrophobic interactions between them;;

mosaic

- b. refers to proteins / glycoproteins embedded in the phospholipid bilayer / AW in a random and scattered arrangement;;

- (b) Phospholipids are a type of lipid. Lipids, in general, are made up of glycerol and fatty acids monomers covalently bonded together. Name the covalent bond and describe the breakage of this bond. [2]

- a. ester bond;; **R! ester**

- b. Addition of 1 water molecule across ester bond via hydrolysis, products of hydrolysis are the hydroxyl group (-OH) in the glycerol molecule and the carboxyl group (-COOH) of a fatty acid;;

[Turn Over

- (c) List four features of cell surface membranes of eukaryotic cells that are **not** visible in Fig. 2.1. and outline their roles in the cell surface membrane. [4]

Feature	Role
a) cholesterol;	✓ Regulate fluidity / OR ✓ increase flexibility and stability of membrane;
b) <u>unsaturated</u> fatty acids; Accept phospholipid tails for fatty acids	✓ Forms kinks which prevents close packing of phospholipids, allowing membranes to remain fluid at low temperatures; (prevents freezing of membranes)
c) carbohydrate chains added to protein(s) / glycoproteins; A! oligosaccharides for carbohydrate chains	✓ cell-to-cell recognition in defense recognition by immune system OR ✓ cell-to-cell adhesion to form tissues OR
d) carbohydrate chains added to lipids / glycolipids;	✓ receptor sites for chemical signals (e.g. hormones) ;
e) channel protein; A! aquaporin	✓ Confer selective permeability, allow specific polar molecules or ions to pass through the membrane;
f) carrier protein;	
g) AVP;;	

Any 4

R! peripheral / extrinsic / integral / intrinsic / transmembrane, proteins
R! attachment to, cytoskeleton / microfilaments

- (d) The inner and outer membrane of the mitochondrion differ in the detail of their membrane components. The inner membrane is also much less permeable than the outer membrane.

Suggest **two** ways in which the structure of the inner membrane is different from that of the outer membrane to produce a **less permeable** inner membrane. [2]

- a. reduced gaps between membrane molecules;;
- b. higher proportion of phospholipids with saturated fatty acids / ora;;
- c. fewer unsaturated fatty acids so, fewer 'kinks' in tails / closer packing;;
- d. higher proportion of cholesterol molecules;;
- e. fewer, channel / carrier / transport, proteins;;
- f. smaller diameter of channels in non-specific channel proteins;;
- g. fewer types of (specific), transport / carrier, proteins;;
- h. AVP;; e.g. fewer, aquaporins / channels for water;;

Any 2

[Total: 10]

3 The central dogma of molecular biology describes the flow of genetic information from DNA to messenger RNA (mRNA) to protein.

(a) A molecule involved in the flow of genetic information from mRNA to protein is transfer RNA (tRNA). Outline the role of tRNA in the production of a polypeptide. [2]

- tRNA attaches to a specific acid and carries it to the ribosome during the elongation phase of translation;;
- where the anti-codon on the incoming tRNA forms hydrogen bonds / ref. to complementary base pairing, with the codon on the mRNA;
- peptide bond formation then takes place between the amino acid attached to this tRNA molecule and the adjacent amino acid in the polypeptide chain;

Each step in the flow of information from DNA to mRNA to protein provides the eukaryotic cell with a potential control point for regulating its functions by adjusting the amount and type of proteins synthesised.

Fig. 3.1 shows the regulation of gene expression of the albumin gene during transcription in eukaryotes. The albumin gene is associated with two control elements and a promoter.

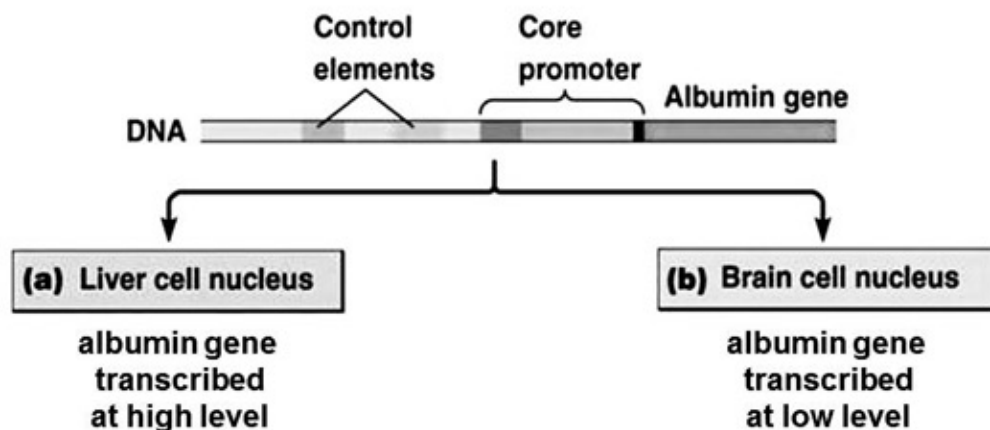


Fig. 3.1

(b) Explain how differential albumin gene expression in liver cells and brain cells is possible. [3]

- Control elements associated with the albumin gene are recognised by specific transcription factors (STFs) which bind to the control elements;;
 - and increase the rate of transcription of the albumin gene by stabilizing the transcription initiation complex (TIC) / higher rate of assembly of the TIC;;
 - These STFs are found in liver cells, therefore albumin gene is transcribed at high levels in liver cells;;
 - Brain cells do not contain these STFs that specifically bind to these two control elements associated with the albumin gene, so there is only a basal / low level of transcription of the albumin gene in brain cells;
- max. 3

[Turn Over

In a normal person, the fragile X mental retardation protein (FMRP), regulates the synthesis of neuron proteins by stopping ribosomal translocation on target mRNAs. Fig. 3.2 shows how

patients with fragile X syndrome (FXS) have non-functional FMRP, resulting in accelerated synaptic protein synthesis that leads to abnormal synaptic function and intellectual disability.

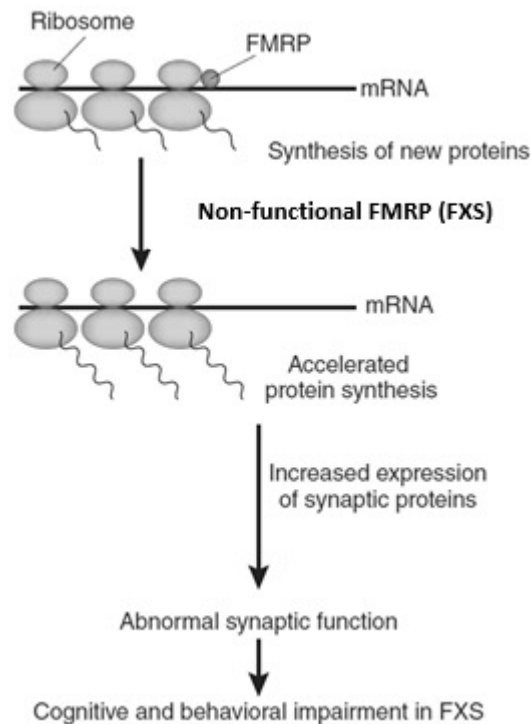


Fig. 3.2

(c) (i) State the level of control by FMRP on synaptic protein expression. [1]

Translational control;;

(ii) Describe one other control mechanism of a similar level as (c)(i). [2]

a. initiation factors;;

b. Activation of initiation factors facilitate the initiation of translation;;

OR

c. binding of regulatory proteins to 5' untranslated regions (5'UTR) of mRNA;;

d. hence preventing initiation of translation / binding of ribosome to mRNA;

OR

e. mRNA degradation / half-life of mRNA;

f. more translation for mRNA with longer poly(A) tails as such mRNA can serve as a template for translation for a longer time;

Research pertaining to gene regulation can contribute significantly to the treatment of genetic diseases. Spinal muscular atrophy (SMA) is a heritable motor neuron disease where patients have insufficient levels of functional survival motor neuron (SMN) protein in their motor neurons and muscle. Genetic studies have shown that all SMA patients have at least one copy of the functional SMN gene at another chromosomal locus (a result of chromosomal duplication), which is not expressed.

Scientists have shown in a study that trichostatin A (TSA), a histone deacetylase inhibitor, caused increased SMN protein levels, improved motor function and survival in mice.

Fig. 3.3 shows some results from the study. Different doses of TSA were injected in the mice and after two hours, muscle tissues were isolated. The levels of acetylated histone H3, histone H4 and SMN mRNA were measured.

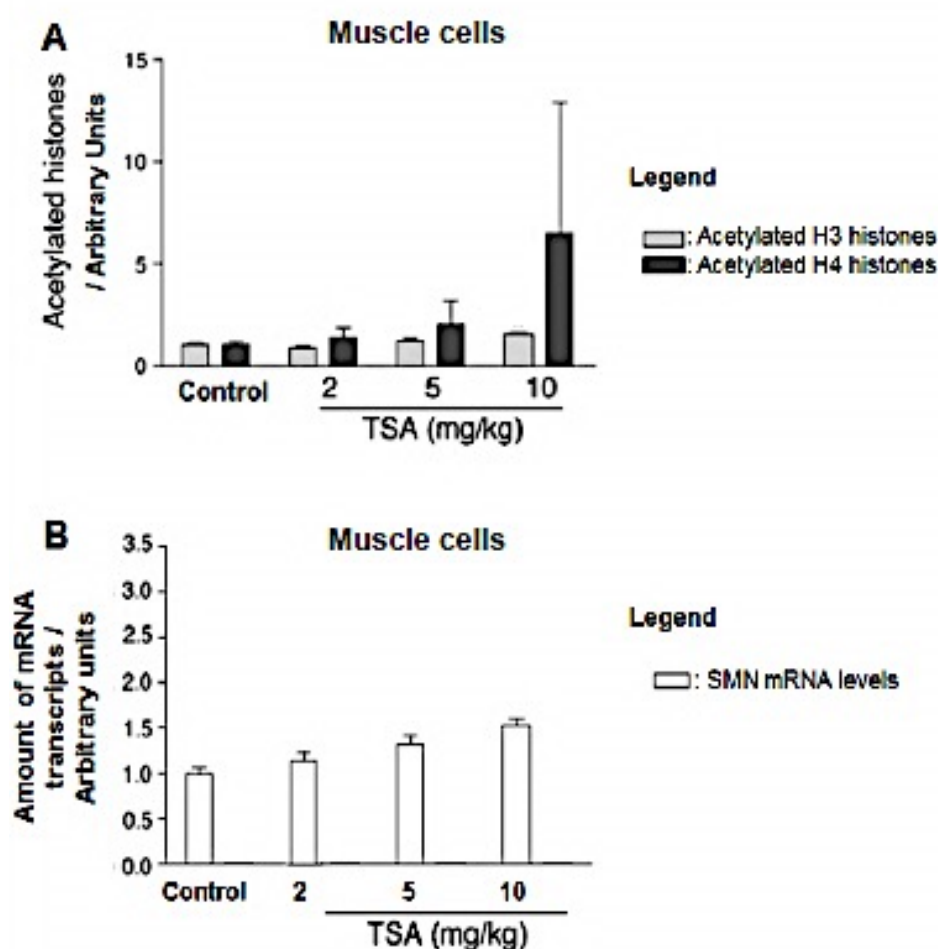


Fig. 3.3

(d) With reference to Fig. 3.3,

- (i) describe the effect of TSA on the amount of SMN mRNA transcripts in muscle cells.
[1]

as amount of TSA increases from 0mg/kg to 10mg/kg, the amount of mRNA transcripts in muscle cells increases from 1.0 a.u. (in the control cells) to 1.5 a.u.;;

[Turn Over

- (ii) explain the effect of an increase in H4 histone acetylation on the regulation of transcription of the SMN gene in muscle cells. [3]
- a. H4 histone tails rich in positively charged amino acid residues (e.g. lysine) interact strongly / form ionic bonds with negatively charged phosphate groups of the DNA backbone;;
 - b. When there is an increase in H4 histone acetylation, addition of the acetyl group neutralises the positive charge on the histone tails;;
 - c. reducing its affinity to DNA, chromatin now has a looser structure;
 - d. and transcription factors now have easier access to the genes in the acetylated region, thus promoting initiation of transcription;
- max. 3

[Total: 12]

- 4 There have been many breakthroughs in stem cell research in the recent years. It has been discovered that stem cells are involved in the replacement of worn-out cells and repair of damaged tissues. Further research is being conducted to better understand the mechanism involved in controlling the behaviour of stem cells in order to better manipulate them to treat various diseases and disorders.

Stem cells undergo cell division to produce genetically identical daughter cells. Fig. 4.1 shows two cells, each at a different stage of cell division.

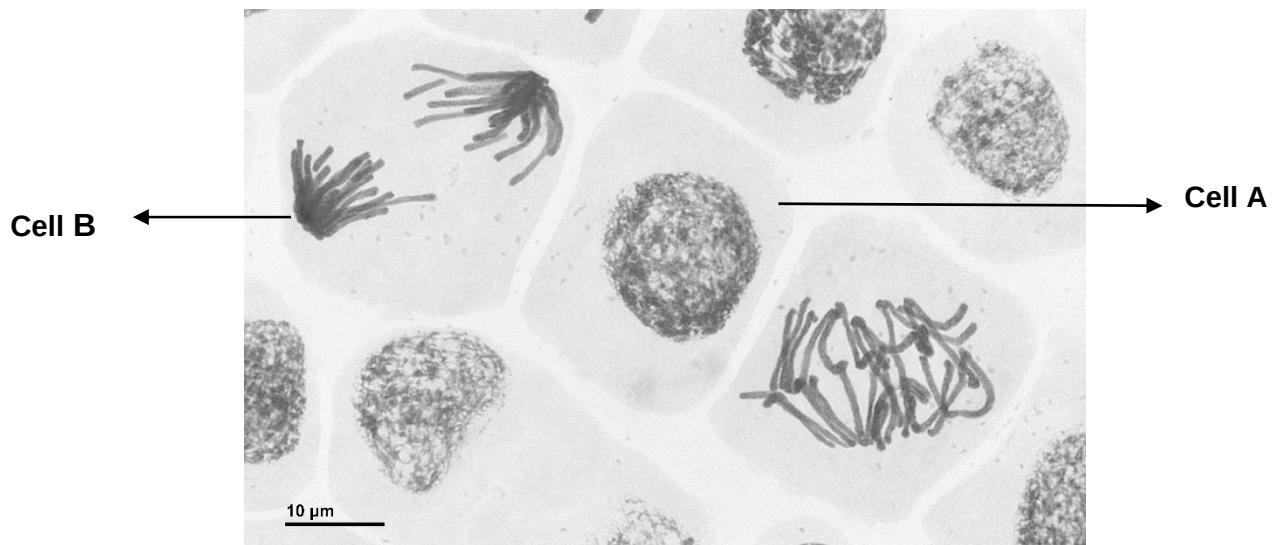


Fig. 4.1

- (a) With reference to Fig. 4.1, state the stages of cell division in Cell A and Cell B. [1]

Cell A **Interphase/Prophase;**

Cell B **Anaphase;**

[Turn over

Fig. 4.2 shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.

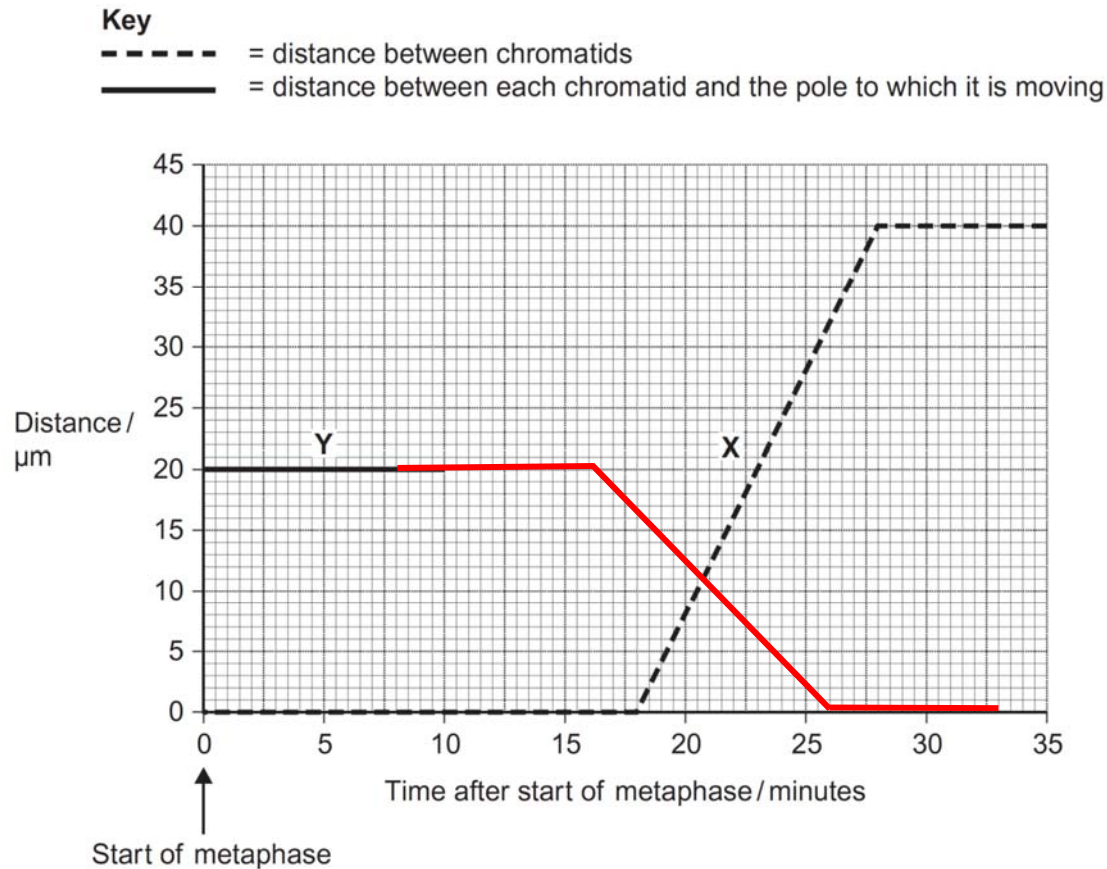


Fig. 4.2

(b) (i) With reference to Fig. 4.2, state the duration of metaphase in the cell. [1]

- **18 min;;**

(ii) Complete line Y on the graph.

[1]

(iii) Account for your answer in (b)(ii). [3]

- Chromosomes align singly at the metaphase plate during metaphase of mitosis OR sister chromatids are attached to microtubules from opposite poles at metaphase;;**
- Sister chromatids start to separate to become daughter chromosomes and migrate towards the opposite poles in anaphase, as shown at 18th min of line X when distance between chromatids starts to increase. Hence distance between chromatid and pole will start to decrease at 18th min;;**
- Distance between chromatids reach a plateau/maximum at 28th min, chromosomes arrived at opposite poles. Hence, distance between chromatid and pole will be minimum at 28th min;;**

The movement of chromatids is dependent on spindle fibres, which are made up of many tubulin subunits. Spindle fibres are lengthened at one end during mitosis by the polymerisation of tubulin subunits through GTP hydrolysis.

(c) Contrast between the structure of tubulin with that of DNA. [2]

Tubulin	DNA
• Tubulin is a polypeptide and hence made up of amino acid subunits;	• DNA is made up of deoxyribonucleotides;
• Subunits by peptide bonds;	• Subunits joined by phosphodiester bonds;
• Globular;	• Helical;
• AVP;	• AVP;

Any 2

[Total: 8]

[Turn over

5 Cancer cells do not heed the normal signals that regulate cell cycle.

(a) Describe the development of cancer as a multi-step process. [3]

- Accumulation of somatic mutations is needed to produce all the changes characteristic of a full-fledged cancer cell;
- gain of function mutations in proto-oncogenes, where only one allele needs to be mutated into an oncogene, and loss of function mutations in tumour suppressor genes (TSGs), where mutations must be in both alleles;;
- gene for telomerase is activated, expression of telomerase in cancer cells removes a natural limit on the number of times the cells can divide;;
- cancer cells escape normal confines of epithelial layer and invade tissue immediately around it, blood vessels formed via angiogenesis to bring oxygen and nutrients to the cancer cells;
- cancer cells enter bloodstream or lymphatic systems, and travel to other tissues / organs, at these new locations, when cancer cells cross epithelium barriers, metastasis has occurred;;

max. 3

Fig. 5.1. shows a cell cycle-inhibiting pathway involving the p53 protein in a normal cell.

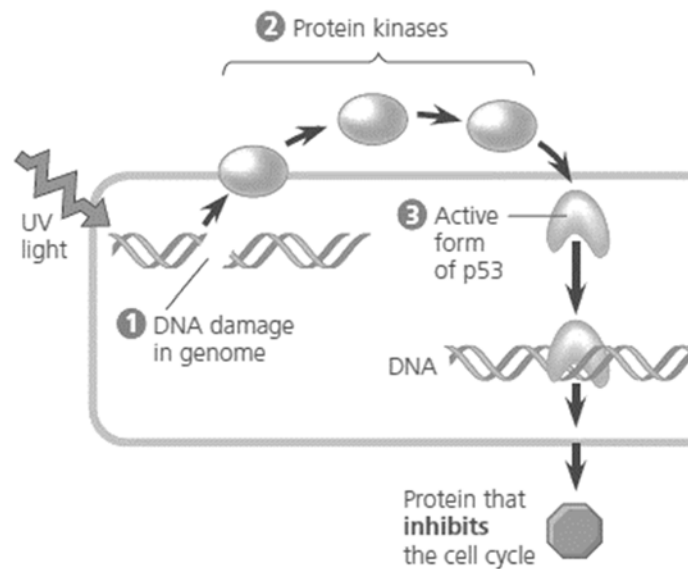


Fig. 5.1

(b) With reference to Fig. 5.1, explain how a missense mutation in p53 protein increases the likelihood of a cell becoming cancerous. [2]

- p53 is a specific transcription factor / activator, in its active form, it binds to DNA to activate the transcription of genes of proteins that inhibit the cell cycle;;
- Missense mutation results in a change in amino acid sequence, resulting in a change in the 3D conformation of the p53 protein, so mutant p53 protein cannot be activated by the protein kinases / cannot bind to the DNA to activate transcription;;
- Protein that inhibits the cell cycle is not synthesized, cell cycle is not inhibited even when there is DNA damage in the genome;

max. 2

(c) Explain why mutations in the p53 gene are considered to be recessive. [2]

- a. Both copies of p53 gene must be mutated / double-hit mutation / both alleles must be knocked out before the tumour suppressor function of p53 is lost;;
- b. If only one allele is mutated, effect of the mutation is masked by the normal dominant (functional) allele and the cell is still able to synthesise sufficient quantities of the p53 protein;;

When a particular retrovirus that does not carry oncogenes infects a particular organism, the amount of mRNA transcribed from a particular proto-oncogene became elevated approximately 20-fold compared with uninfected individuals.

(d) Suggest an explanation for the above observation. [2]

- a. Retrovirus may pick up a copy of host's proto-oncogene and integrate it into its own viral genome;;
- b. new virus oncogene is expressed at abnormal levels under the control of the viral promoter;;

OR

- c. Retrovirus may integrate as a provirus (by chance) near one of the cell's proto-oncogenes;;
- d. strong promoter / enhancer in the provirus stimulates high-level of transcription of proto-oncogene;;

[Total: 9]

[Turn over

- 6 The fruit fly, *Drosophila melanogaster*, is widely used in genetic research. It has many phenotypic variants in features such as body colour, wing shape and eye colour.

Two variations from the normal-winged, grey-bodied phenotype are:

- vestigial (very short) wings, coded for by the recessive allele of the gene **N/n**
- ebony (black) body colour, coded for by the recessive allele of the gene **G/g**.

- (a) Using the symbols given, state the possible genotypes of normal-winged, grey-bodied fruit flies. [2]

NNGG
NNGg
NnGG
NnGg

4 correct = 2 marks

2/3 correct = 1 mark

R! letters besides N, n, G, g

- (b) Describe how you would determine the genotype of a normal-winged, grey-bodied fly.[3]

- a. **Perform a test cross, cross fly with, vestigial wing and ebony body fly OR double / homozygous, recessive fly / nngg fly;;**
 b. **if some offspring have vestigial wing and / or ebony body, genotype is heterozygous;;**

A! if, some offspring have recessive trait / not all offspring have dominant trait, genotype is heterozygous

- c. **if offspring all have normal wing and / or grey body genotype is homozygous;;**

A! if offspring all have dominant trait genotype is homozygous

A! short for vestigial and black for ebony throughout

- (c) One of the genes for eye colour is carried on the X chromosome. This gene has different alleles coding for

- red eyes
- brown eyes
- white eyes.

The allele for red eyes (R) is dominant to the allele for brown eyes (b) and dominant to the allele for white eyes (w). The allele for brown eyes is dominant to that for white eyes.

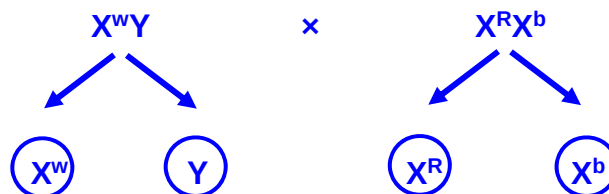
Using these symbols, draw a genetic diagram to show how a cross between a white-eyed male fruit fly with a red-eyed female fruit fly will produce male and female offspring that are either red-eyed or brown-eyed.

Genetic diagram

Parental phenotype white-eyed male × red-eyed female

Parental genotype

Gametes



Offspring genotype and phenotype

	X^R	X^b
X^w	$X^R X^w$ red-eyed female	$X^b X^w$ brown-eyed female
Y	$X^R Y$ red-eyed male	$X^b Y$ brown-eyed male

- a) Correct parental genotypes;;
- b) Correct gametes with circles and arrows;;
- c) Correct offspring genotype;;
- d) Correct offspring phenotype;;

wrong symbols = 0

superscripts R, b, w on Y chromosome = 0

no X and Y = Max 1

ecf alleles written as subscripts not superscripts = Max 3

ecf superscript R written as small r = Max 3

ecf superscript b/w written as big B/W = Max 3

[4]

[Turn over

- (d) The “eyeless” gene is a master control gene that directs the growth and development of the eyes. In flies homozygous for the mutant allele of “eyeless”, the body is not instructed to make eyes during development, resulting in a blind fruit fly as shown in Fig. 6.1. Thus when ‘eyeless’ gene is mutated, there is no expression of the gene for eye colour even though the fly may carry two copies of the gene for eye colour.

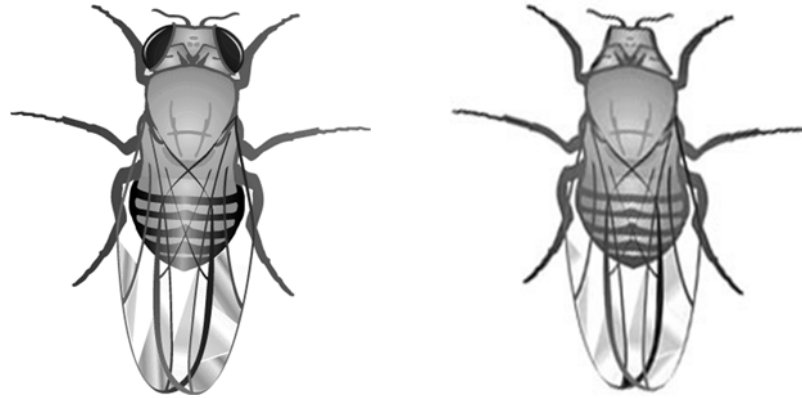


Fig. 6.1 Wild type fruit fly (left), blind fruit fly (right)

State the name for this type of interaction between “eyeless” gene and the gene for eye colour. [1]

(Recessive) epistasis;;

- (e) A statistical test was performed to investigate whether there was a significant difference in the mean lifespan between wild type and blind fruit flies. A summary of the results is shown in Table 6.1.

Table 6.1

Mean lifespan of fruit flies / days		Significance of difference	Total sample size
Wild type	Blind		
45	35	$p > 0.05$	50

Comment on what the results show. [2]

- a. degree of freedom is 48, probability that any difference in means is due to chance is more than 0.05 / 5%;;
 b. thus, there is no significant difference in the mean lifespan between wild type and blind fruit flies, H_0 not rejected;;

[Total: 12]

- 7 The uptake of radioactively-labelled carbon dioxide in chloroplast was investigated.
- Three tubes, each containing different components of chloroplasts, were exposed to light.
- The results of investigation are shown in Table 7.1.

Table 7.1

Tube	Contents	Uptake of radioactively labelled carbon dioxide/ counts per minute
A	stroma and grana	3,500
B	stroma, ATP and reduced NADP	3,400
C	stroma	300

- (a) Name the substance that combines with carbon dioxide in chloroplast. [1]

RuBP / ribulose biphosphate;;

- (b) Explain why the results in tube B are similar to those in tube A. [3]

(in tube B)

- Grana is site of light-dependent reactions;;**
- where ATP and reduced NADP are formed;;**
- thus light-independent stage / Calvin cycle can proceed at a similar rate to tube A;;**

- (c) Explain why the uptake in tube C was less than the uptake in tube B. [3]

(without ATP and reduced NADP)

- less / no, G3P converted to TP ;;**
- less / no, RuBP / ribulose biphosphate, can be regenerated;;**
- light-independent stage / Calvin cycle, occurs at a low rate / cannot proceed;;**

[Turn Over

- (d) Cyanobacteria are prokaryotic organisms. Chloroplasts are thought to have evolved from cyanobacteria that became incorporated into larger cells. Experiments show that free-living cyanobacteria can adapt to environmental signals in the same way as chloroplasts. Fig. 7.1 shows the absorption spectra of two species of cyanobacteria

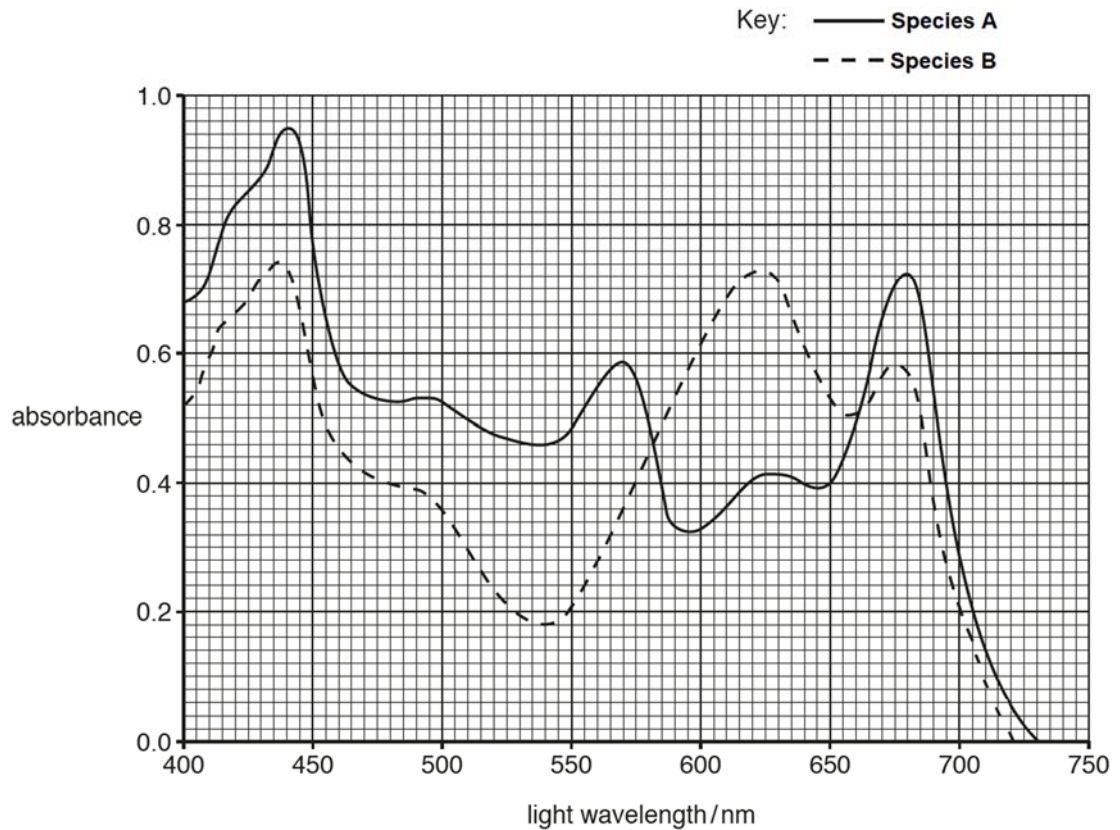


Fig. 7.1

With reference to Fig. 7.1, describe the differences in absorption spectra between the two species of cyanobacteria. [3]

- Species A has greater absorbance than Species B in 400 – 580(585)nm and 660(665) – (720) 730nm range;; ora
- Species A has lower absorbance than Species B in 580(585)nm – 660 (665)nm range;; ora
- Absorbance for Species A peaked at 440nm while absorbance for Species B peaked at 440nm and 625nm;;
- Species A is able to absorb wavelengths between 720 to 730nm while Species B is unable to absorb in this range;;

Max 3

[Total: 10]

8 A recent study of the house mouse, *Mus musculus*, on the island of Madeira resulted in the following observations.

- There are six distinct populations.
- The mice are associated with human settlements.
- The populations are located in different valleys separated by steep mountains.
- Each population has a different diploid number of chromosomes.

As a result of these observations it has been suggested that speciation is taking place.

Fig. 8.1 is a map of Madeira showing the distribution of the six populations.

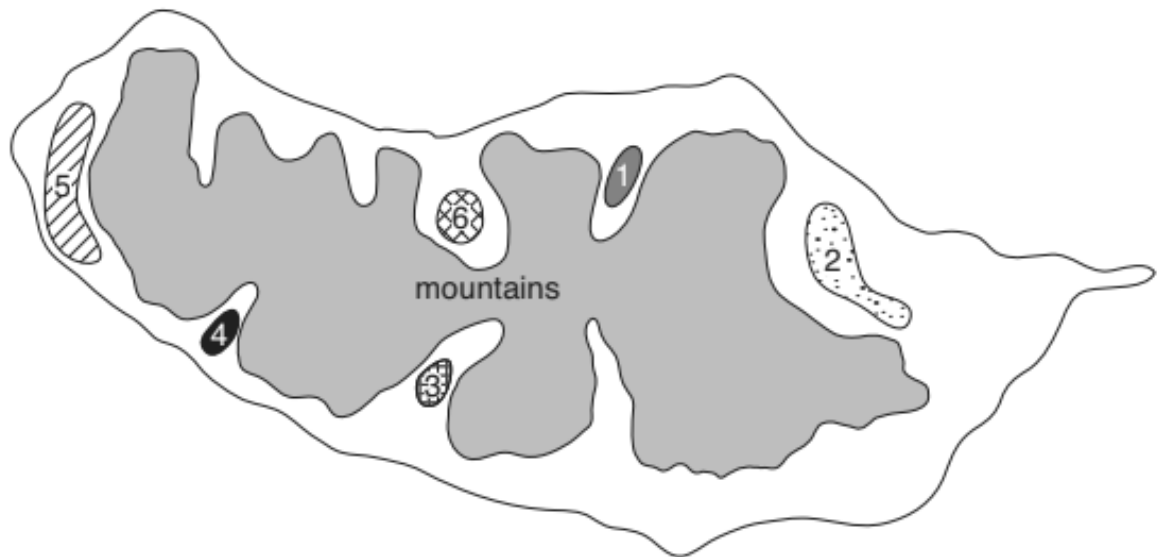


Fig. 8.1

(a) Using the information in Fig. 8.1, state the likely isolating mechanism and the type of speciation taking place. [2]

Isolating mechanism: **geographical/mountains/physical barrier;;**

Type of speciation: **allopatric;;**

(b) Explain how speciation is occurring in the house mouse populations of Madeira. [5]

- Mouse populations separated by mountains/ ref to geographical isolation;;
- No breeding/gene flow, between populations;;
- Mutations occur and accumulates in the population;;
- Different selection pressures/different (environmental) conditions;;
- Genetic change; e.g. different alleles selected for/ change in allele frequency/ change in gene pool/ advantageous alleles passed on;;
- (results in) different chromosome numbers;;
- Genetic drift/founder's effect;;
- (different populations ultimately) cannot interbreed; R! different species;;

[Turn over

(c) Explain the likely outcome of individuals from two separate populations being mated in captivity. [2]

- b. Due to different chromosome number / diploid number;;
- c. No homologous chromosomes, hence no pairing takes place. Meiosis cannot take place, thus no gametes will be produced by offspring, hence viable but infertile;;

Based on earlier research, the native house mouse species and subspecies were distinguished based on a number of phenotypic differences such as length of the tail and fur colour. However, with modern technology, researchers have been using molecular methods such as direct DNA and amino acid comparison to further determine the phylogeny between the native house mouse species.

(d) Explain why molecular homology is better than anatomical homology in determining evolutionary relationship between species of native house mouse. [2]

- a. Similarity in anatomical features could be due to convergent evolution;
- b. Study of anatomical homology is not possible between morphologically different species;
- c. Using molecular method, organisms can be compared even if they are morphologically very different & all organisms have certain molecular traits in common e.g. rRNA sequences or certain fundamental problems;
- d. Using morphological features as a benchmark is subjective and non-quantifiable;
- e. Molecular data is quantifiable and objective. Nuclei acid and amino acid sequence data are precise and easy to quantify, hence allows an objective assessment of evolutionary relationships;
- f. Fossils obtained from ancestral species may be incomplete thus comparative study of anatomy is not possible;
- g. Compare molecular divergence of ancestral species with incomplete/no fossil record with that found in other lineages with more complete fossil fuels;

Max 2 marks

[Total: 11]

9 Fig. 9.1 shows the glucagon signalling pathway.

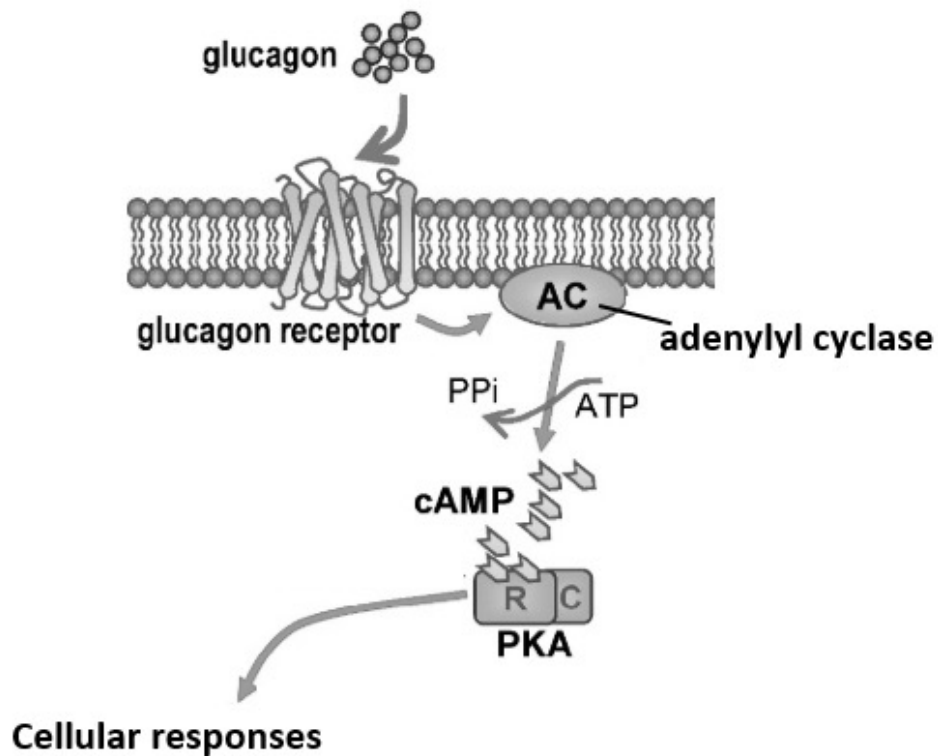


Fig. 9.1

- (a) In humans, in which types of cells are glucagon receptors mainly found? [1]
- Liver cells;;**
A! Kidney cells
- (b) Describe how the glucagon receptor transmits information from the external environment to activate adenylyl cyclase inside of the cell. [3]
- The ligand, glucagon, binds to the GPCR, which is linked to a G-protein and triggers conformational changes in the GPCR, exposing the cytoplasmic domain of the GPCR;;
 - G protein binds to GPCR and exchanges GDP for GTP, and GTP-bound G protein becomes active;;
 - G protein dissociates from the receptor, (moves along the cell surface membrane), and binds to adenylyl cyclase to activate it.

[Turn over

(c) Describe two cellular responses resulting from the cell signalling pathway shown in Fig. 9.1. [2]

- a. Glycogenolysis / conversion of stored glycogen to glucose, releasing this glucose into the bloodstream;;
- b. Stops the liver cells from consuming glucose, helping more glucose to remain in the bloodstream;;
- c. Gluconeogenesis / production of glucose from amino acids or other sources;;

max. 2

(d) In signal transduction pathways, how can the response of the target cell to a hormone be amplified? [2]

- a. Amplification occurs at the signal transduction stage, where the binding of one ligand molecule (at the receptor) activates relay molecules which in turn activate an increasing number of downstream relay molecules;;
- b. As such, the number of activated relay molecules is much higher than the preceding step;;
- c. Signal amplification can be achieved via phosphorylation cascades, e.g. activation or the production of second messengers;;
- d. Signal amplification requires the activated relay molecules or second messengers to remain in an active form long enough to activate a high number of downstream molecules;;

max. 2

[Total: 8]

10 Our immune system needs to respond promptly to infections by pathogens. These pathogens are usually bacteria and viruses. Both our innate and adaptive immune systems are essential in helping our bodies fight against bacterial pathogens.

(a) Name one type of cell in the innate immune system and state its role in eradicating bacterial infections. [1]

- a. (tissue-resident) macrophage / dendritic cell engulfs bacteria by phagocytosis, using hydrolytic enzymes in lysosomes to break down the bacterial cells;;
- b. (tissue-resident) macrophage / dendritic cell (migrates to the lymph node to) activate the adaptive immune system / T helper cells;;
- c. circulating monocytes and neutrophils are phagocytes are easily recruited to sites of infection quickly to help eliminate the microbial infection;;

max. 1

Cholera is an infectious disease caused by the bacterium, *Vibrio cholerae*. Most people who have recovered from cholera rarely become ill again from the disease. In these people, antibodies have been identified that will bind either to the cholera toxin, or to the bacterium flagellum, or to the main bacterial cell.

(b) Explain why the antibodies are different, each one specific to its target.[3]

- a. Toxin, flagellum and main bacterial cell are recognised by the adaptive immune system as different antigens;;
- b. Each different antigen activates a specific clone of B-lymphocytes, leading to the secretion of the type of antibodies specific to that antigen;;
- c. different antibodies have different F_{ab} / variable regions for binding to different antigens;;

OR

- d. different antibodies have different specificity / 3D complementarity to different antigens;;

Fig. 10.1 shows part of the process of the production of a heavy chain polypeptide for an antibody. At the top of the diagram is the chromosomal arrangement found in an immature B cell.

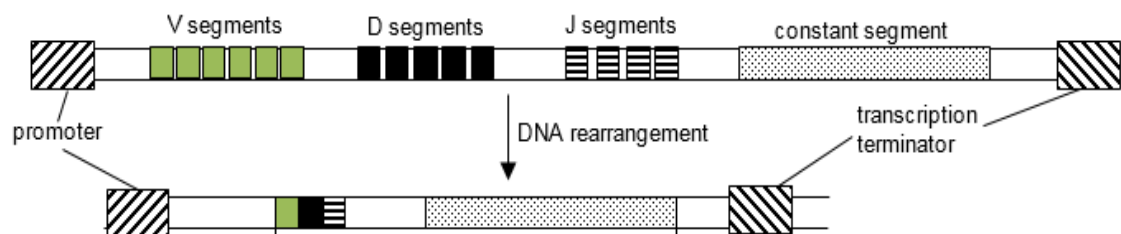


Fig 10.1

[Turn over

- (c) Identify the process represented in Fig. 10.1 and state the enzyme involved in the process. [1]

Process is somatic recombination, enzyme is VDJ recombinase;;

Vaccinations for some diseases are available. The development of the influenza vaccine is a dynamic research process as frequent genetic changes in the surface antigens of the virus renders each vaccine less effective over time. However, another mechanism of genetic change that results in flu pandemics can also occur irregularly and unpredictably. This occurs when two different strains of the influenza virus infect the same host cell simultaneously.

- (d) Suggest why the influenza virus can undergo this irregular and unpredictable mechanism of genetic change. [1]

a. Influenza virus can undergo antigenic shift because of genetic recombination of its 8 genome segments / segmented genome;;

[Total: 6]

- 11** Surface ocean carbon dioxide concentration can be determined by recording the concentration of carbon dioxide, in a closed volume of air that was circulated with a constantly renewed supply of water obtained two to three meters below the surface of the ocean.

Fig. 11.1 are graphs showing the changes in concentration of carbon dioxide in the air and changes in pH in the oceans of Bermuda and Hawaii from 1990 to 2010.

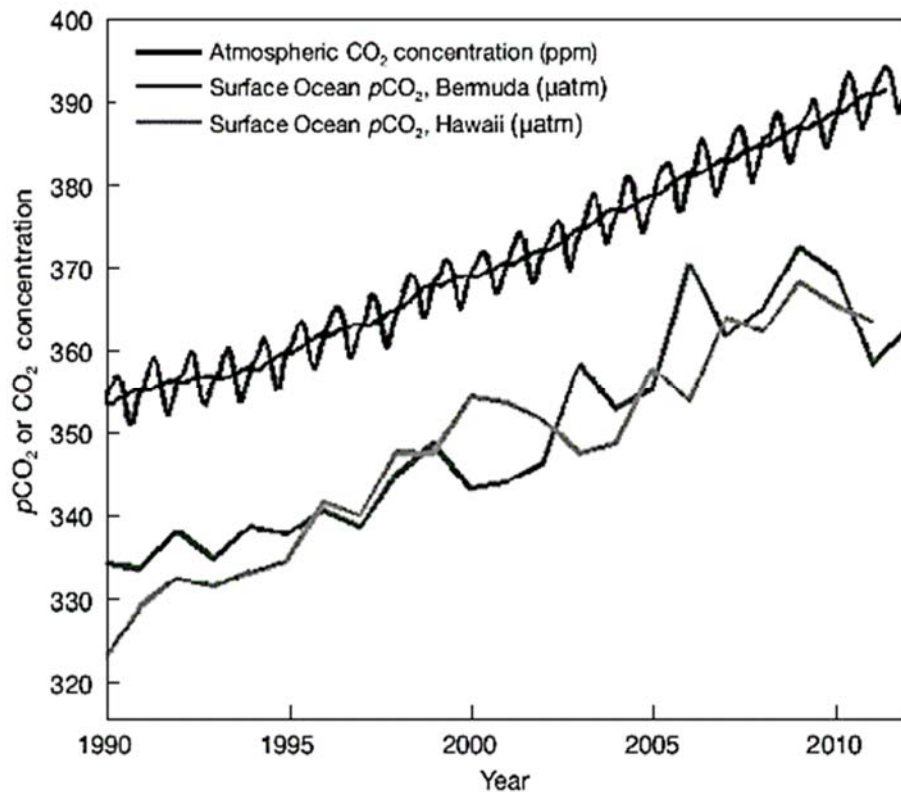


Fig. 11.1

- (a) Briefly explain two human activities that have resulted in increased emission of carbon dioxide into the atmosphere. [2]
- b. increasing energy usage which requires the combustion of fossil fuels to generate electricity;;**
 - c. Deforestation results in the removal of forests which are the carbon sink. Burning the forest releases the stored organic carbon back into atmosphere as CO₂;;**
 - d. Increasing consumption of meat means that carbon dioxide is released indirectly by the agriculture industry rather than the consumers into the atmosphere;;**

Max 2

Increasing ocean surface carbon dioxide concentration also has negative impacts on coral reefs.

Coral reefs are at risk of damage by human activities. A study was conducted to see the effects of climate change on coral reefs.

[Turn over

Coral reef sites were subjected to two different environmental conditions i.e. exposed site and sheltered site. Coral reefs in exposed site was exposed to climate change environmental conditions. Coral reefs in the sheltered site was exposed to normal environmental conditions.

Table 11.1 shows coral cover area at exposed and sheltered sites.

Table 11.1

Experimental site		Area of healthy coral reef/m ²	Average area of healthy coral reef/m ²
Exposed Site	Site 1	120	
	Site 2	100	
	Site 3	150	
Sheltered Site	Site 1	82	
	Site 2	75	
	Site 3	69	

With reference to Table 11.1,

- (a) Complete Table 11.1 by calculating the average area of healthy coral reef in exposed and sheltered site. [1]

Show your working

$$x_1 = (120 + 100 + 150)/3 \\ = 123.3\text{m}^2$$

$$x_2 = (82 + 75 + 69)/3 \\ = 75.3\text{m}^2$$

- (b) Explain two ways how climate change damages coral reefs.[2]
- increase in green house gasses emission in atmosphere traps heat in atmosphere warms atmospheric temp & absorbed by water bodies/ocean;;
 - at higher water temp, increased photosynthesis rate of zooxanthellae leading to excess product which is toxic;;
 - this damages coral causing coral polyp to expel zooxanthellae which results in coral being bleached;;
 - ocean absorbs increased amt of carbon dioxide in the air causes ocean acidification/drop in pH of the ocean;;

Max 2

[Total: 5]

END OF PAPER